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Roll No

EC-5001-CBGS

B.E. V Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Electromagnetic Field Theory

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) State and prove Gauss's Law.
b) Define gradient, curl and divergence with suitable examples. Write down their physical significance.
2. a) Explain Coulomb's Law.
b) Find the Equation for Energy Density in the Electrostatic Field.
3. a) Derive the Poisson's and Laplace's Equation.
b) Find the Boundary conditions between two magnetic materials.
4. a) Obtain boundary conditions for magnetic field
b) Derive the expression for the magnetic flux density produced by an infinite this long wire carrying current I at a distance R.

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5.
 - a) Explain Maxwell's equation in integral and differential forms.
 - b) Derive and explain Helmholtz wave equation. Write down the properties of plane waves.
6.
 - a) Discuss about plane waves in lossy medium and good conducting medium.
 - b) Discuss about Interference of two plane waves traveling at oblique directions.
7. Discuss about reflection and refraction of plane waves at dielectric media and conducting surfaces.
8. Write short notes on (any three)
 - a) Magnetic vector potential for sources in free space
 - b) Transmission line analogy
 - c) Uniqueness theorem
 - d) Frequency dispersive propagation
